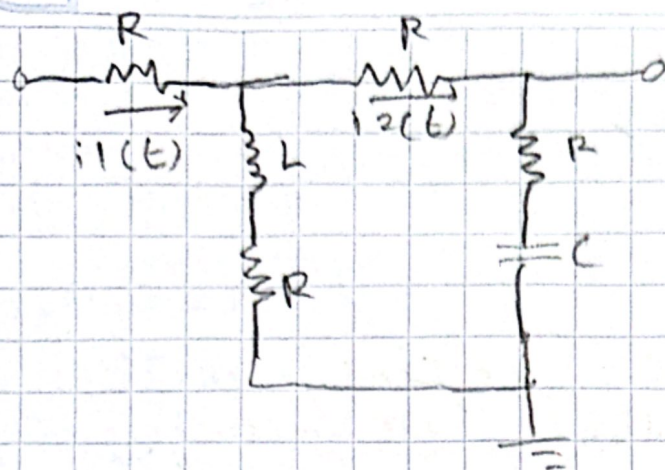




Ecuaciones principales

$$v_c(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)] = R i_2(t) + R \int i_2(t) dt$$

$$v_2(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_c(s) = R I_1(s) + L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)]$$

$$L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] = R I_2(s) + R \frac{I_2(s)}{s}$$

$$V_2(s) = R I_2(s) + \frac{I_2(s)}{s}$$

procedimiento algebraico.
Función de transferencia

$$\frac{V_S(s)}{V_E(s)} =$$

Agrupar corrientes

$I_1(s)$ e $I_2(s)$

Expresar el voltaje de entrada en función de $I_2(s)$

Despejar $I_1(s)$ de la segunda ecuación principal

$$V_S(s) = \left(R + \frac{1}{Cs} \right) I_2(s) = \left(\frac{CRs + 1}{Cs} \right) I_2(s)$$

$$LS I_1(s) - LS I_2(s) + RI_1(s) - RI_2(s) = RI_2(s) + \frac{I_2(s)}{Cs}$$

$$LS I_1(s) + RI_1(s) = RI_2(s) + RI_2(s) + \frac{I_2(s)}{Cs} + LS I_2(s) + RI_2(s)$$

$$(LS + R) I_1(s) = \left(LS + \frac{1}{Cs} + 3R \right) I_2(s)$$

$$I_1(s) = \frac{1}{LS + R} \left(LS + \frac{1}{Cs} + 3R \right) I_2(s)$$

$$I_1(s) = \frac{CLS^2 + 3(CRs + 1) I_2(s)}{CLS^2 + CRs}$$

$$V_E(s) = RI_1(s) + LS I_1(s) - LS I_2(s) + RI_1(s) - RI_2(s)$$

$$V_E(s) = (LS + 2R) I_1(s) - (LS + R) I_2(s)$$

$$V_E(s) = \left[(LS + 2R) \frac{CLS^2 + 3(CRs + 1) I_2(s)}{CLS^2 + CRs} - (LS + R) I_2(s) \right]$$

$$V_E(s) = \left[\frac{(LS + 2R)(CLS^2 + 3(CRs + 1)) - (LS + R)(CLS^2 + CRs)}{CLS^2 + CRs} \right] I_2(s)$$

$$sCR^2 + R$$

$$V(s) = \frac{[(s+2R)(CLs^2 + 3CRs + 1) - (s+R)(CLs^2 + CRs)]}{CLs^2 + CRs} I_2(s)$$

$$V(s) = \frac{CRs + 1}{CLs^2 + (3CR + L)s + 2R} I_2(s)$$

$$V(s) = \frac{CRs + 1}{CLs^2 + (3CR + L)s + 2R}$$

$$= \frac{CRs + 1}{CLs^2 + (3CR + L)s + 2R}$$

$$= \frac{(s+R)(CRs + 1)s + R}{3CLs^2 + (3CR + L)s + 2R}$$

Error en

Error en estado estacionario cuando se aplica un escalón unitario de entrada

$$e(s) = \lim_{s \rightarrow 0} \frac{1}{s} \left(\frac{1 - CLs^2 - (3CR + L)s + R}{3CLs^2 + (3CR + L)s + 2R} \right) = \frac{R}{2R}$$

$$e(t) = 0.5V$$

Estabilidad en lazo abierto

Se calculan los polos del sistema (raíces del denominador)

$$R = 10 \text{ K}\Omega$$

$$L = 15 \text{ mH}$$

$$C = 220 \text{ nF}$$

$$\lambda_1 = -1.818181848$$

$$\lambda_2 = -1.111111093$$

0.5

transitorio estado estacionario

El sistema es estable con una respuesta sobreamortiguada
 Modelo de ecuaciones integro-diferenciales

A partir de las ecuaciones principales del sistema, se
 despejan las variables independientes

$$v_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$L i_1(t) = [v_s(t) + R i_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt}] \frac{1}{2R}$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + R i_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$v_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$R = 2.2 \times 10^3$$

$$L = 15 \times 10^{-6} \text{ mH}$$

$$C = 220 \times 10^{-9} \text{ F}$$

$$N_s = \frac{(LRs^2 + (CR^2 + L)s + R)}{3CLR s^2 + (5CR^2 + L)s + 2R} = \frac{3.26 \times 10^{-6} s^2 + 1064.8 s + 2200}{2.178 \times 10^{-5} s^2 + 5324 s + 4400}$$